Adani University

Bachelor of Technology

Subject Name: Data Communication & Computer Networks

Subject Code: EICCI24203 Sem: 4th Date: 23-03-2026

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EXPERIMENT: 10

OBJECTIVE:

To implement HTTP protocol of application layer in Cisco Packet Tracer.

SOFTWARE USED:

CISCO PACKAGE TRACER, DESKTOP

THEORY:

Advantages

- Simple and easy to implement
- Widely supported by all web browsers
- Allows cross-platform communication between any client and server that understand the protocol.
- Can transfer various data types (text, images, video) by indicating the content type in the headers.

Disadvantages

- Not secure (data is transmitted in plain text)
- Since it doesn't remember previous requests, managing user sessions (like logging in) requires extra overhead like cookies.
- Can experience delays due to the underlying TCP connection setup (three-way handshake) required before sending HTTP requests.

CODE:

Not applicable

PRACTICAL:

1.With 6 computers:

Figure 1A :- Basic Diagram

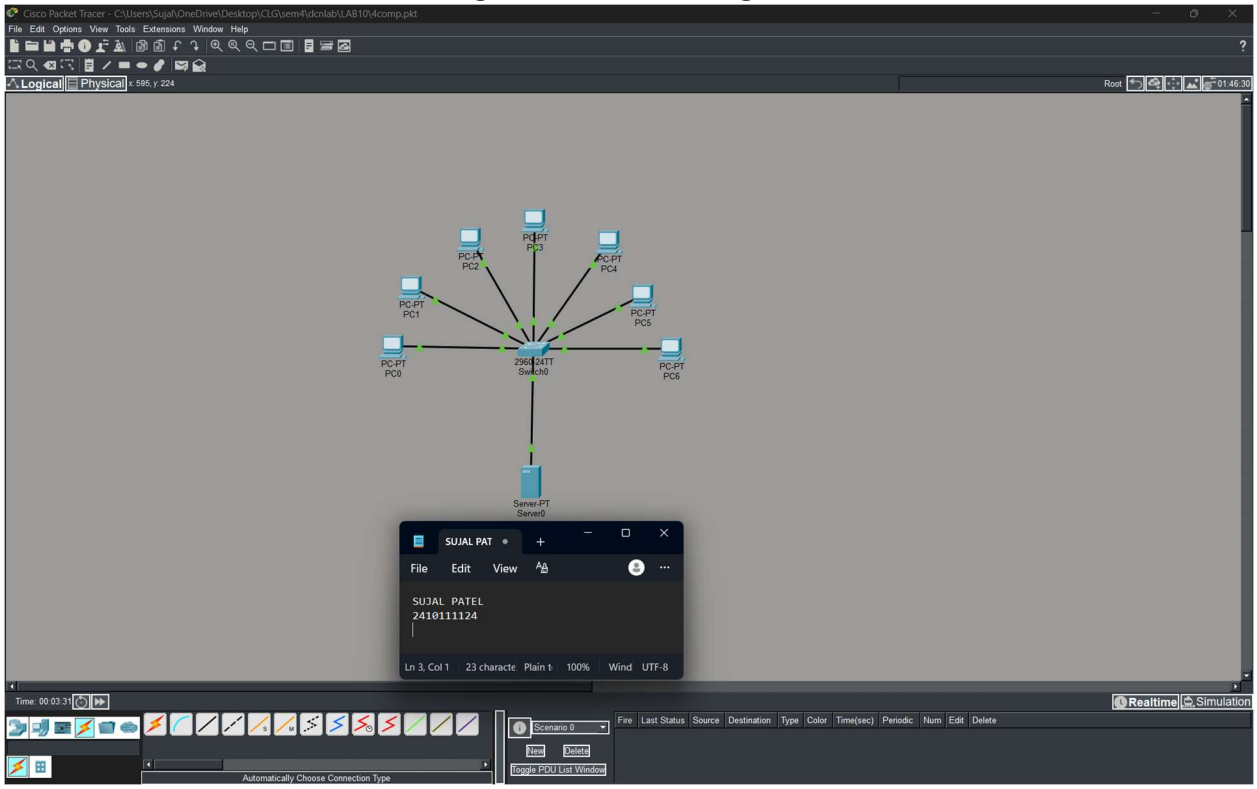


Figure 1B :- Simulation Diagram

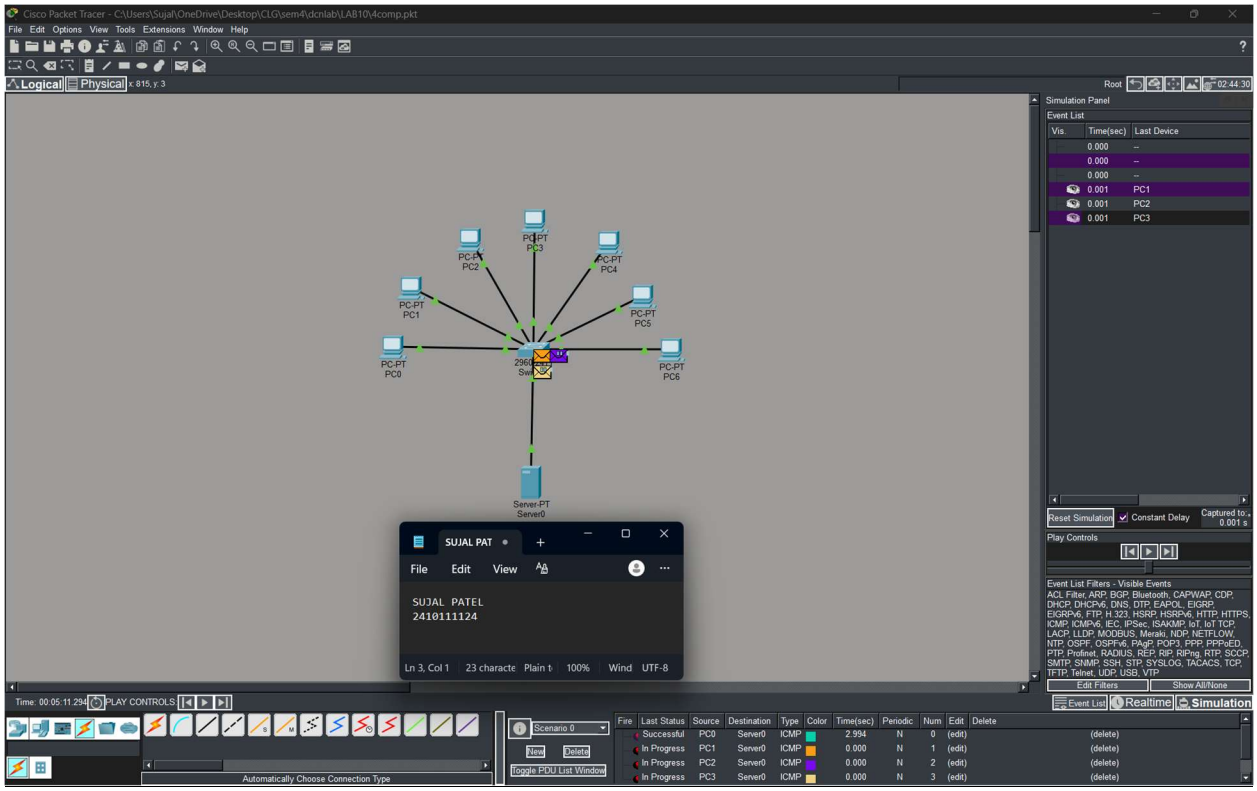
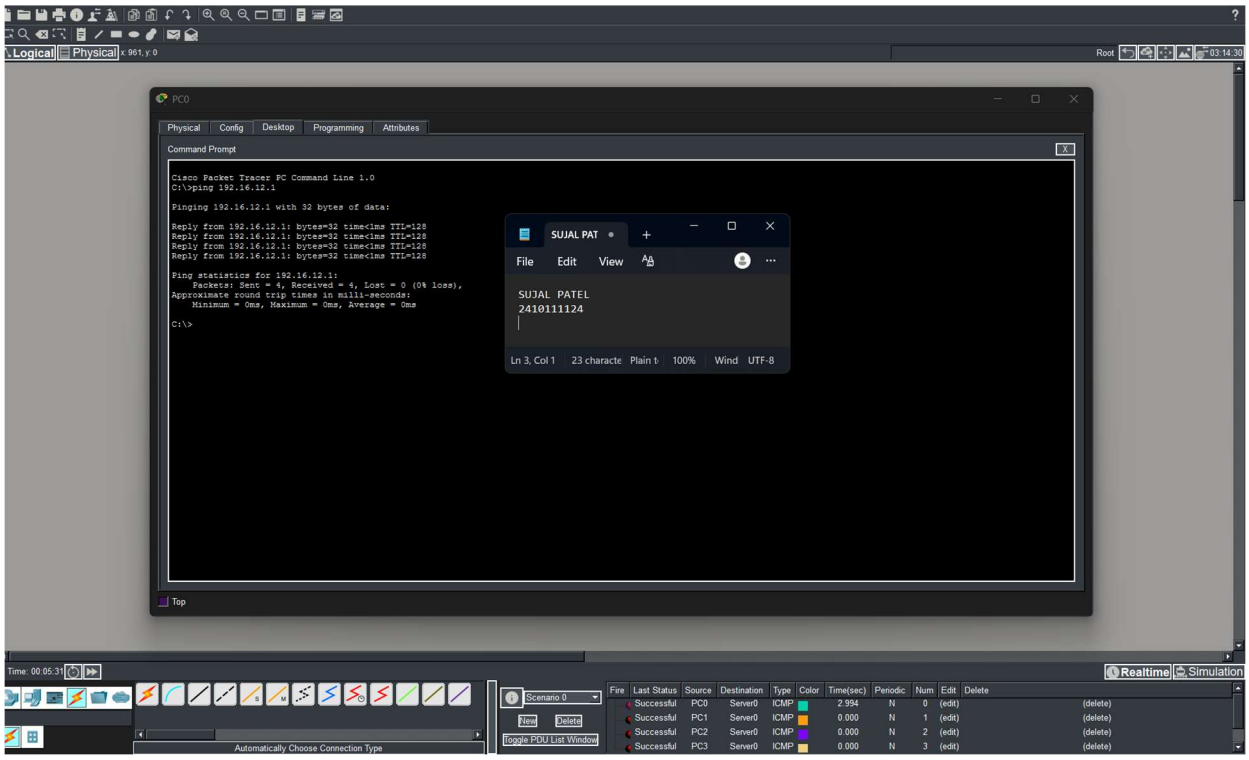
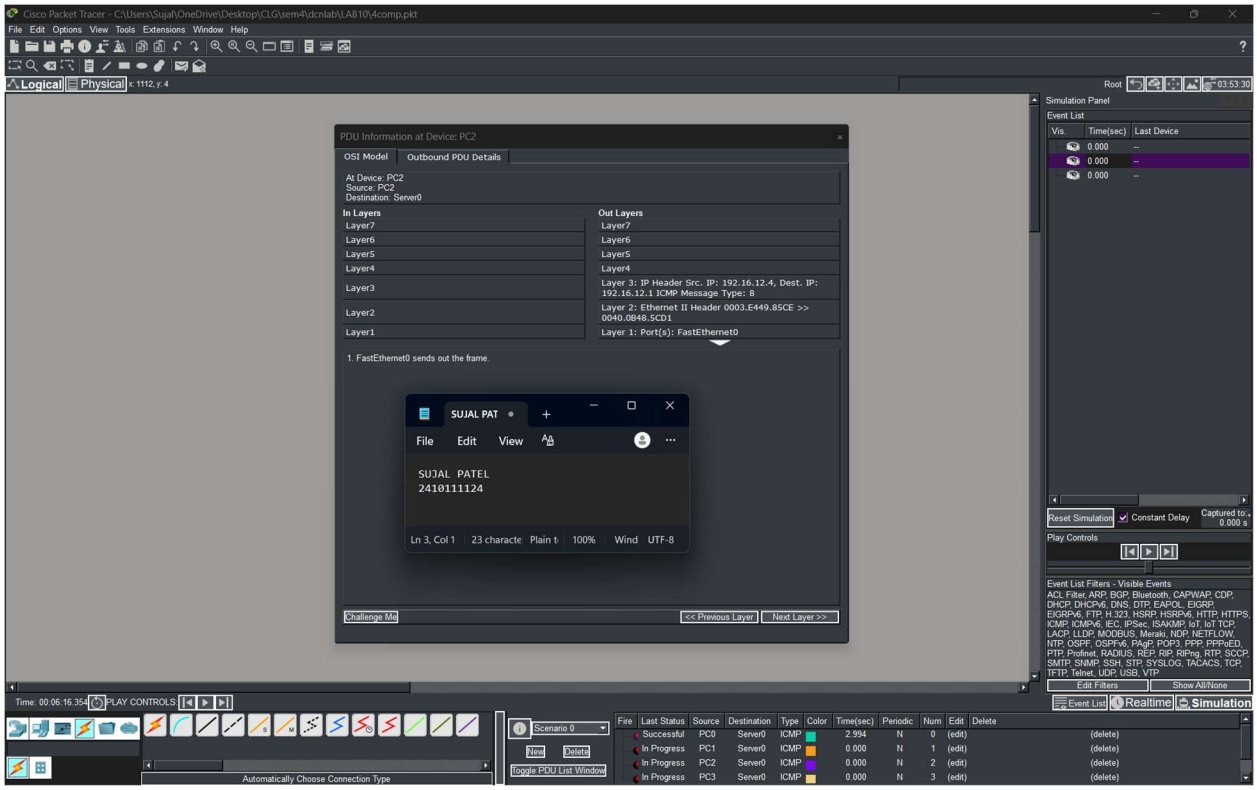
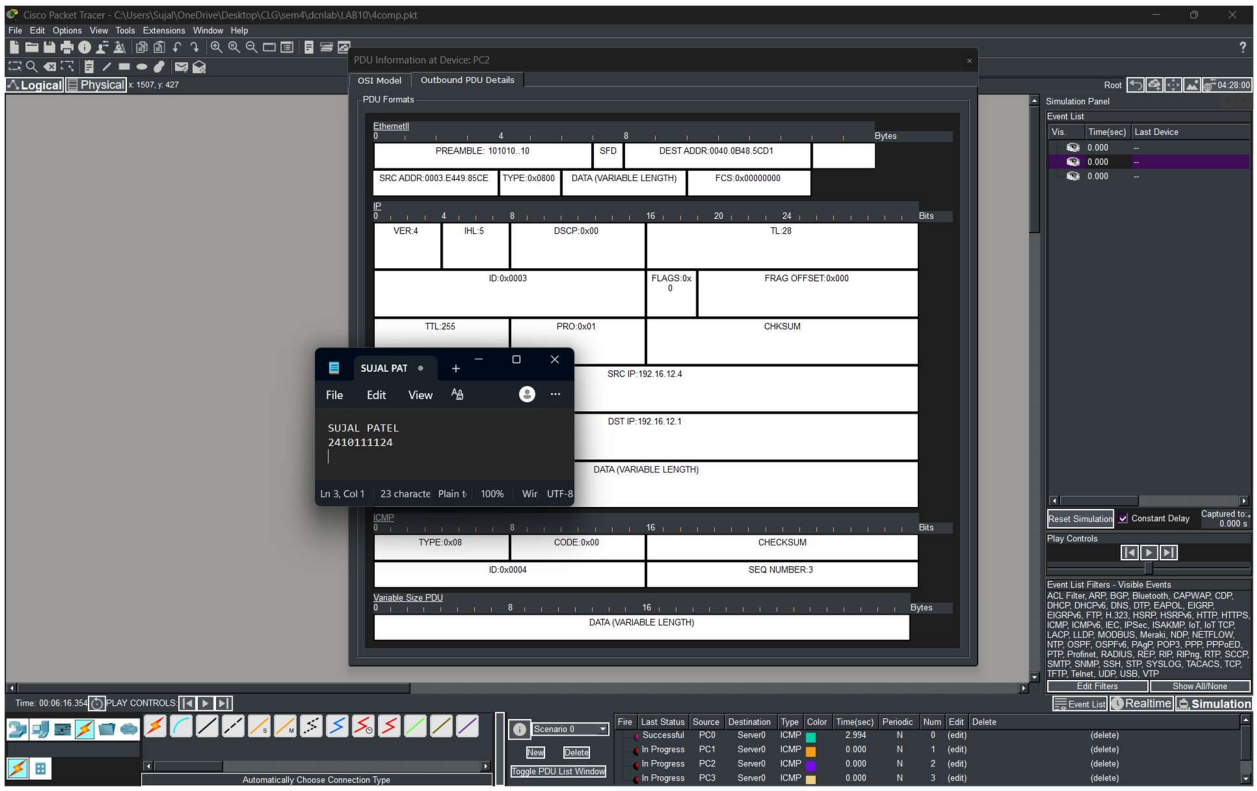
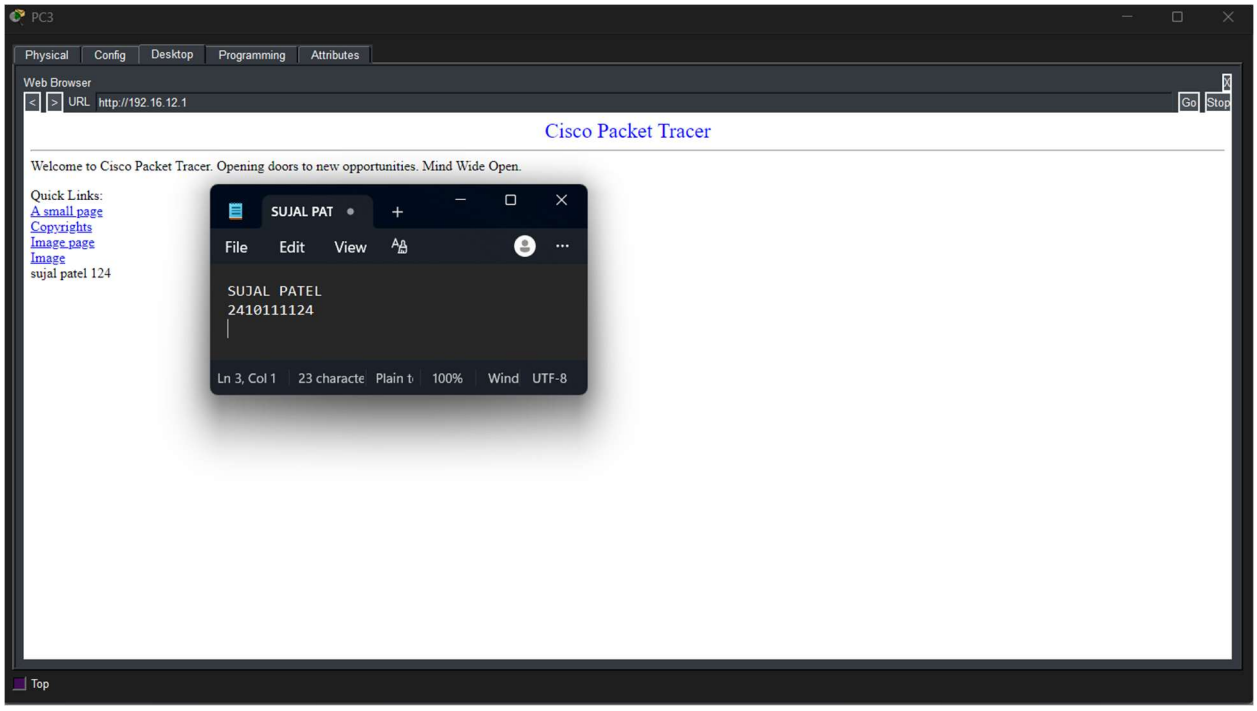


Figure 1C :- Analysis of message received




PDU Information at Device: PC2
OSI Model
Outbound PDU Details
At Device: PC2
Source: PC2
Destination: Server0
In Layers
Out Layers
Layer7
Layer6
Layer5
Layer4
Layer3
Layer2
Layer1
Layer 3: IP Header Src. IP: 192.16.12.4, Dest. IP: 192.16.12.1 ICMP Message Type: 8
Layer 2: Ethernet II Header 0003.E449.85CE >> 0040.0B48.5CD1
Layer 1: Port(s): FastEthernet0
1. FastEthernet0 sends out the frame.
SUJAL PAT
File Edit View
SUJAL PATEL
2410111124
Challenge Me
<< Previous Layer
Next Layer >>
Simulation Panel
Event List
Vis. Time(sec) Last Device
Reset Simulation
Constant Delay
Play Controls
Event List Filters - Visible Events
Edit Filters
Show All/None
Realtime
Simulation
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete
Successful PC0 Server0 ICMP 2.994 N 0 (edit) (delete)
In Progress PC1 Server0 ICMP 0.000 N 1 (edit) (delete)
In Progress PC2 Server0 ICMP 0.000 N 2 (edit) (delete)
In Progress PC3 Server0 ICMP 0.000 N 3 (edit) (delete)


PDU Information at Device: PC2
OSI Model
Outbound PDU Details
PDU Formats
Ethernetll
PREAMBLE: 101010..10
SFD
DEST ADDR:0040.0B48.5CD1
SRC ADDR:0003.E449.85CE
TYPE:0x0800
DATA (VARIABLE LENGTH)
FCS:0x00000000
IP
VER:4
IHL:5
DSCP:0x00
TL:28
ID:0x0003
FLAGS:0x0
FRAG OFFSET:0x000
TTL:255
PRO:0x01
CHKSUM
SRC IP:192.16.12.4
DST IP:192.16.12.1
DATA (VARIABLE LENGTH)
ICMP
TYPE:0x08
CODE:0x00
CHECKSUM
ID:0x0004
SEQ NUMBER:3
Variable Size PDU
DATA (VARIABLE LENGTH)
SUJAL PAT
File Edit View
SUJAL PATEL
2410111124
Simulation Panel
Event List
Reset Simulation
Constant Delay
Play Controls
Event List Filters - Visible Events
Edit Filters
Show All/None
Realtime
Simulation
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete
Successful PC0 Server0 ICMP 2.994 N 0 (edit) (delete)
In Progress PC1 Server0 ICMP 0.000 N 1 (edit) (delete)
In Progress PC2 Server0 ICMP 0.000 N 2 (edit) (delete)
In Progress PC3 Server0 ICMP 0.000 N 3 (edit) (delete)



PROCEDURE:

1. Cisco Packet Tracer was launched and a new project was created.
2. The required devices such as one server, one PC, and one switch were placed in the workspace.
3. The PC and server were connected to the switch through cables.
4. Appropriate IP addresses were assigned to the PC and the server within the same network.
5. The server was opened and the HTTP service was enabled from the services tab.
6. A simple web page was created or verified on the server.
7. The network was allowed to stabilize in real-time mode.
8. The PC was opened and the web browser application was selected.
9. The IP address of the server was entered in the browser URL field.
10. Successful loading of the web page confirmed HTTP communication.

RESULTS:

The HTTP protocol was successfully implemented. The PC was able to access the web page hosted on the server using a web browser, confirming proper communication at the application layer.

POSSIBLE SOURCES OF ERROR:

- Incorrect IP address configuration
- HTTP service not enabled on server
- Wrong cable connection
- Devices not properly connected to the switch
- Typing wrong URL in browser

CONCLUSION:

The experiment successfully demonstrated the implementation of the HTTP protocol in Cisco Packet Tracer. It verified effective communication between client and server, enhancing understanding of application layer protocols and practical networking concepts.

LEARNING OUTCOME:

1. Understanding of HTTP protocol
2. Knowledge of client-server communication
3. Practical exposure to application layer protocols

PRECAUTIONS:

1. Ensure correct IP configuration for all device
2. Always turn ON HTTP service on the server
3. Enter correct URL in browser